1.9 Rewriting Equations and Formulas

Name _____ Period ____ Date _____

Algebra I 1.9 · Student sheet

I can…

- solve a literal equation for one of its variables.
- rewrite a formula for a given variable.
- use a rewritten formula to solve real-life problems.

Vocabulary · fill in the missing word

Literal equation — an equation with two or more _____

Formula — a literal equation that relates _____, such as A = lw

Solve for a variable — get that variable alone using inverse _____

Part 1 · Same steps

Number equation	Literal equation
3x + 5 = 20 (solve for x)	3x + b = c (solve for x)
4x − 2 = 18 (solve for x)	ax − d = e (solve for x)

What do you notice? _____

Part 2 · Notes and examples (fill in each note, then work the example)

Literal equation: A literal equation has two or more _____.

Solving for a variable: To solve for a variable, treat the other letters like _____.

Example 1

Solve 3x + 2y = 8 for y.

Example 2

Solve y = 5x − 3 for x.

Example: Rewrite A = ½bh for h: h = _____

Example 3

Solve each formula for the given variable.

a. $A = \frac{1}{2}bh$ for h b. $P = 2l + 2w$ for w

Example 4

Solve $F = (\frac{9}{5})C + 32$ for C. Then find the Celsius temperature when it is 68°F.

Part 3 · Partner practice

Partner 1

Solve $6x - 3y = 12$ for y.

Partner 2

Solve $V = lwh$ for w.